

Umakant Kulkarni

Contact

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Current Address

New Providence, NJ 07974 USA

PROFILE SUMMARY

Ph.D. research scientist specializing in AI, machine learning, and scalable intelligent network systems, with experience building end-to-end LLM- and reinforcement learning-driven agentic AI systems. Expertise includes custom transformer attention mechanisms, parameter-efficient fine-tuning, curriculum learning, and distributed model training and serving. First-author publications in leading IEEE and ACM venues, with a published AI-based systems patent in the United States, Germany, and China.

CORE AI/ML SKILLS

- **LLMs & Fine-Tuning:** LoRA / PEFT, continued pre-training, instruction/discriminative fine-tuning, custom attention layers, tokenization (SentencePiece, tiktoken)
- **Curriculum & Training:** Multi-stage curriculum learning, domain adaptation, composite losses (hinge-margin, differentiable AUC), evidential/uncertainty modeling
- **Reinforcement Learning (RL):** Markov Decision Process, Deep Q-Networks (DDQN), recurrent (LSTM) policies, reward design, simulation-based training
- **Frameworks:** PyTorch, HuggingFace Transformers / PEFT / Accelerate / TRL, DeepSpeed, TensorFlow, Keras, scikit-learn
- **Inference & Serving:** vLLM, FlashInfer, llama.cpp, Triton, TensorRT, quantization (bitsandbytes, llmcompressor), PyTorch Distributed (DDP, FSDP, RPC), Ray, FastAPI, Gradio
- **RAG & Data:** FAISS, BM25, sentence-transformers, NumPy, Pandas, data pipelines, Weights & Biases, Hydra
- **Infra:** CUDA, multi-GPU (NVIDIA), Jetson Orin, Docker, Kubernetes, AWS, Azure, Terraform

SELECTED AI/ML PROJECTS

Janus — Dual-Mask Attention Transformer (LLM)

PyTorch · LLaMA · PEFT/LoRA · DDP · FAISS

- Designed a novel dual-mask attention layer on LLaMA, splitting attention heads into causal (temporal) and semantic (field-level) groups in one forward pass.
- Fine-tuned with LoRA/PEFT plus custom classification and evidential heads; composite loss combining hinge-margin ranking and a differentiable AUC surrogate.
- Built a curriculum learning pipeline: domain-adaptive pre-training → normal-behavior characterization → discriminative fine-tuning.
- Orchestrated multi-GPU distributed training (DDP, fp16) via Accelerate.

Maestro — Deep RL Agent for Adaptive Control

Reinforcement Learning · DDQN · LSTM · PO-MDP

- Formulated resource allocation task as a PO-MDP and trained a Double-DQN agent with an LSTM to act under partial observability from noisy telemetry.
- Designed a probabilistic reward model from black-box signals (no client instrumentation); built a simulation environment to accelerate RL training.

Split LLM Inference — Device ↔ Edge

PyTorch Distributed RPC · vLLM · llama.cpp · Serving API

- Built a distributed split-inference engine partitioning LLaMA-3.1-8B across device/edge with PyTorch RPC; implemented custom split protocol for hidden state, KV-cache, and masks per decode step.
- Engineered a framework to serve and benchmark LLMs/VLMs with vLLM, llama.cpp, low-precision quantization, and OpenAI-compatible APIs across edge/data-center hardware.

TECHNICAL SKILLS

- OS: Windows, Linux, UNIX, iOS
- Languages: C/C++, Golang, Python, Java, MATLAB, JavaScript | Scripting: Bash, Tcl/Tk, Perl
- Tools/Technologies: Helm, Istio, Fluentd, Prometheus, Grafana, Kiali, MongoDB, Jaeger, Kibana.

EDUCATION

Ph.D., Computer Science
Purdue University, West Lafayette, USA Jan 2021 - Dec 2025
GPA 4.0

Master of Science, Computer Systems Networking
Northeastern University, Boston, USA Sep 2015 - May 2017
GPA 3.71

Bachelor of Engineering, Electronics and Telecommunication
University of Pune, India Aug 2011 - May 2015
First Class with Distinction

RESEARCH AND PROFESSIONAL EXPERIENCE

Nokia Bell Labs, Murray Hill, NJ USA
Research Scientist Jan 2026 – Present

- Conducting research at the intersection of cellular networks (5G/6G), distributed systems, network security, and AI/ML to advance next-generation mobile network systems.
- Designing resilient distributed 6G core network architectures in collaboration with KDDI, focusing on core-network decomposition, distributed state management, fault tolerance, and service continuity.

Affirmed Networks/Microsoft, Acton MA USA
Principal Software Engineer Sep 2018 – Dec 2020

- Developed stateless, containerized telecommunication software based on 5G standards for core network functions AMF, SMF, UPF, N3IWF, PCF and UDM that can be deployed via Docker and/or Kubernetes to achieve scalability, resiliency and redundancy.
- Implemented Istio service-mesh features into 5G NFs for observability, security and traffic management.
- Designed a build environment to define, install and upgrade applications on Kubernetes using Helm.
- Developed DPDK containerized UE-gNB simulator for testing different data-plane functionality.
- Built pyATS-enabled automation infrastructure and integrated it with JIRA and Jenkins for simulating CI/CD model of cloud-native microservice architecture.

Software Quality Assurance Engineer Jun 2017 – Aug 2018

- Contributed towards architectural development and quality assurance for Virtualized ePDG (evolved Packet Data Gateway), ‘Security-Gateway’ and T-WAG (Trusted - Wireless Access Gateway) network functions from 4G Evolved Packet Core.
- Automated call-flows related to AAA, PCRF and OFCS Servers, GTP-Path Management, X.509 Security Certificates over S2x, SWm, STa and SWu interfaces, in Tcl and Python.

TEACHING EXPERIENCE

Purdue University, West Lafayette, IN USA Jan 2021 – Dec 2021

- Graduate Teaching Assistant for the course ‘Computer Architecture (CS 250)’

Northeastern University, Boston, MA USA Jan 2017 – May 2017

- Graduate Teaching Assistant for the course ‘Telecommunication Architecture and Systems (TELE 5320)’ for 2 sections.

INTERNSHIPS

InterDigital, Conshohocken PA USA

Research Intern

May 2025 – Aug 2025

- Analyzed, designed, and implemented next generation 6G mobile networks.
- Developed 6G networking protocols, services, and solutions to enable application verticals (GenAI).

Hewlett Packard Enterprise, Milpitas CA USA

Research Associate Intern

May 2022 – Dec 2023

- Characterized QoE of volumetric video streaming over enterprise Wi-Fi networks, analyzing the impact of Wi-Fi control parameters and comparing QoE with traditional 2D video applications.
- Proposed an adaptive method for Wi-Fi resource allocation to maximize the overall system QoE and QoE fairness.

Affirmed Networks, Inc. Acton MA USA

SQA Intern

May 2016 – Jan 2017

- Laid the foundation for ePDG and T-WAG automation infrastructure; Built the ‘smoke’ as well as ‘sanity’ frameworks to automate the end-to-end workflow.
- Modified open-source libraries and configured StrongSwan module onto a linux box as well as Raspberry Pi to support different Wi-Fi calling features over IPsec.

Thuse Electronics Pvt. Ltd., Pune India

Engineering Intern

Nov 2013 – Dec 2013

- Worked on the processes transformer manufacturing, PCB design, PCB assembling in line production, quality and rework management, process quality control, wave soldering on RoHS and non RoHS sections, automatic optical inspection and surface mount technology.

CERTIFICATIONS

Build Large Language Models (LLMs) from Scratch

Vizuara Technologies Private Limited · Issued Oct 2025 · Credential ID 0cd0775996b71249

Responsible Conduct of Research (RCR) Training - Faculty, Postdoctoral, and Graduate Students

CITI Program · Issued Oct 2025 · Expires Oct 2029 · Credential ID 66250359

Certified Telecommunication Engineer (Reg. No. – S13PUN128)

Bharat Sanchar Nigam Limited (BSNL), Govt. of India

- Platinum Level - Mobile Communication, IP Networking and Cyber Security Jul 2014
- Gold Level - Broadband Technology, Optical Fiber Technology Jan 2014
- Silver Level - Digital Switching and Transmission System, Telecom Support Infrastructure Jul 2013

OPEN SOURCE CONTRIBUTIONS

- Scapy: <https://github.com/secdev/scapy>
- Open5GS: <https://github.com/open5gs/open5gs>
- P4 Utils: <https://github.com/nsg-ethz/p4-utils>
- P4 Tutorials: <https://github.com/p4lang/tutorials>
- GenAI-TMF: <https://github.com/InterDigitalInc/GenAI-TMF>

PATENTS

- QoE-aware Dynamic Resource Allocation
Shivang Aggarwal, **Umakant Kulkarni**, Khaled Diab, Lianjie Cao, Faraz Ahmed, Puneet Sharma
Published in USA, Germany and China, April 2026

PUBLICATIONS

- Elastic and Energy-Efficient Management of Mobile Core Control Plane with In-Process WASM Workers - *Submitted at IEEE CNSM, July 2026*
- Modeling Latency of Split Autoregressive Decoding in Cellular Edge Systems
Submitted at Edge Intelligence Workshop at ACM/IEEE Symposium on Edge Computing, July 2026
- Janus: A Dual-Mask Attention Transformer for Log-based Anomaly Detection in Cellular Networks
Umakant Kulkarni, Sonia Fahmy
Proc. ACM Meas. Anal. Comput. Syst. 10, 1, Article 14, (SIGMETRICS'26), 36pp., March 2026.
- RUN: A Case for Cross-Layer Networked Virtual Reality
Yufeng Chen, **Umakant Kulkarni**, Voicu Popescu, Sonia Fahmy
Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), 10 pp., October 2025.
- Maestro: QoE-Aware Dynamic Resource Allocation in Wi-Fi Networks
Umakant Kulkarni, Khaled Diab, Lianjie Cao, Faraz Ahmed, Shivang Aggarwal, Puneet Sharma, Sonia Fahmy
Proceedings of the ACM on Networking, Volume 3, Issue CoNEXT1, Article 4, 24 pp., March 2025
- Securing the Cloud-Native 5G Control Plane
Umakant Kulkarni, Sonia Fahmy
IEEE MILCOM Workshop on Secure FutureG Wireless Communications and Networked Systems, 6 pp., October 2024
- Understanding the Impact of Wi-Fi Configuration on Volumetric Video Streaming Applications
Umakant Kulkarni, Khaled Diab, Shivang Aggarwal, Lianjie Cao, Faraz Ahmed, Puneet Sharma, Sonia Fahmy
ACM SIGCOMM Workshop on Emerging Multimedia Systems (EMS), 6 pp., September 2023
- Toward QoE-based Routing Path Selection
Umakant Kulkarni, Yufeng Chen, Patrick Melampy, Sonia Fahmy
IEEE International Conference on High Performance Switching and Routing (HPSR), 6 pp., June 2023
- Slicure5G: Secure Slicing for 5G (Poster)
Umakant Kulkarni and Sonia Fahmy
USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2 pp. April 2023
- Towards A Low-Cost Stateless 5G Core (Poster)
Umakant Kulkarni, Amit Sheoran, Sonia Fahmy
IEEE International Symposium on Local and Metropolitan Area Networks, 2 pp., July 2022
- The Cost of Stateless Network Functions in 5G
Umakant Kulkarni, Amit Sheoran, Sonia Fahmy
ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS), 7 pp., December 2021

FELLOWSHIPS AND AWARDS

- Elected to membership in The Honor Society of Phi Kappa Phi in January 2025. Membership is by invitation only and recognizes academic excellence among the top 10% of graduate students.
- Received employee merit recognition award under the “High-Profile Projects” category for achieving significant milestones in “Securing 5G for Military Operations” research project at Purdue University
- 3rd Prize in Best-in-class 2022 Summer Interns Project Fair organized by Hewlett Packard Enterprise
- Received “Outstanding Graduate Teaching Assistant Award” for significant accomplishments and highest ratings in teaching and mentoring students at College of Engineering, Northeastern University.
- Awarded by KVPY Fellowship (English: Young Scientist Incentive Plan) with All India Rank – 5. This research fellowship was given by The IISc, Bangalore and Department of Science and Technology.
- Received Best Paper/Research/Poster award at seven regional conferences, India (2011-2015).

PROFESSIONAL SERVICE

- *ACM CoNEXT 2026* — Artifact Evaluation Committee
- *ACM CAIS 2026* — Artifact Evaluation Committee
- *USENIX NSDI 2026* — Artifact Evaluation Committee
- *ACM SIGCOMM 2025* — Artifact Evaluation Committee
- *ACM IMC 2025* — Shadow Technical Program Committee
- **Ad hoc journal reviews:**
 - *IEEE Internet Computing*
 - *IEEE Transactions on Networking*
 - *IEEE Transactions on Network Science and Engineering*
 - *IEEE Transactions on Dependable and Secure Computing*

WEB PROFILES

Homepage: <https://umakantkulkarni.github.io>

GitHub: <https://github.com/UmakantKulkarni/>

LinkedIn: <https://www.linkedin.com/in/umakantkulkarni/>

Google Scholar: <https://scholar.google.com/citations?hl=en&user=1v3d0A0AAAAJ>

REFERENCES

- Prof. Sonia Fahmy
Professor of Computer Science, Purdue University
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- Dr. Nakjung Choi
Department Head - Mobile Network Systems, Nokia Bell Labs
Email: nakjung.choi@nokia-bell-labs.com
- Mark Libby
System Software Manager, Boston Dynamics
Email: mark@mwlibby.net
- Prof. Peter O'Reilly
Director, Telecommunication Networks program, Northeastern University
Email: p.oreilly@northeastern.edu